Philip S. Boonstra

Associate Professor

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Education

PhD, Biostatistics, University of Michigan (Ann Arbor)	2009 – 2012
MS, Biostatistics, University of Michigan (Ann Arbor)	2007 – 2009
BA, Mathematics and Political Science, Calvin University (formerly Calvin College) (Grand Rapids, MI)	2002 – 2006

Professional Affiliations

Associate Professor, with tenure (Dept. of Biostatistics, University of Michigan)	2020 – present
Research Associate Professor (Dept. of Biostatistics, University of Michigan)	2019 – 2020
Research Assistant Professor (Dept. of Biostatistics, University of Michigan)	2013 – 2019
Affiliated Faculty (Center for Cancer Biostatistics, University of Michigan)	2013 – present
Affiliate Member, Cancer Control and Population Sciences Program (University of Michigan Rogel Cancer Center)	2016 – present
Graduate Student Research Assistant (Dept. of Biostatistics, UM)	Intermittent, 2009 - 2012
Graduate Student Research Assistant (Kidney and Epidemiology Cost Center, UM) (Chronic Kidney Disease Surveillance Team, UM Group)	Summers 2008, 2009
Summer Research Assistant (Dept. of Mathematics and Statistics, Calvin College)	Summers 2004, 2006

Honors & Fellowships

Rackham Predoctoral Fellowship (University of Michigan)	2011 - 2012
Rackham Conference Travel Grant (UM)	2010, 2011, 2012
Rackham Regents' Fellowship (UM)	2009 - 2011
NIH Predoctoral Research in the Biosciences Training Grant (UM)	2007 - 2009
1st Place, Michigan Autumn Take Home (MATH) Challenge (Calvin College)	2005
Dean's List (GPA $>$ 3.5/4.0) (Calvin)	2002 - 2006
Presidential Scholarship (Calvin)	2002 - 2006

Teaching

Traditional

Public Health 345* (University of Michigan; 2 credit hours) Winters, 2025-2026
Undergraduate course teaching the principles and application of data visualization with applications to public health datasets
* Newly developed course

Biostatistics 699 (University of Michigan; 4 credit hours) Winters, 2019-2026
Capstone course for 2nd-year MS students/1st year PhD students

Biostatistics 619 (UM; 3 credit hours) Winter 2018

Biostatistics 619 (2 credit hours) Falls 2014-2016
Graduate course on the design, conduct, and analysis of clinical trials

Biostatistics 644 (UM; 2 credit hours) Fall 2015
Graduate course on statistical methods for cancer research

Teaching Assistant / Grader (Calvin College) 2004 - 2005
for undergraduate courses in calculus, discrete mathematics, and linear algebra

Non-traditional

Data Science for Health Research Specialization (Coursera; Massive Open Online Course Series) Launched Sep 2023
Asynchronous three-course series for health data science learners (co-develped/co-taught with Professor Bhramar Mukherjee)
`https://www.coursera.org/specializations/data-science-for-health-research`

Biostatistics 591* (University of Michigan; 2 credit hours) Springs, 2020-2023
Introduction to R programming course for online MPH students (hosted on Coursera)
* Newly developed course

Big Data Summer Institute (UM) Summer 2017
Faculty mentor for 6 week undergraduate research program

Publications

1. Gutierrez MG, Wang C, Inamdar K, Stone MB, Ehyaee V, Venkataraman G, **Boonstra PS**, Murga-Zamalloa C, and Wilcox RA "GATA-3 identifies a distinct subset of T-cell prolymphocytic leukemia (T-PLL) and is associated with CCR4 expression" In press at *Blood Neoplasia*
2. Conte ML, **Boonstra PS**, Caverly TJ, Fishstrom A, Flynn AJ, Taksler GB, and Friedman CP. "The potential for a benefit-estimating algorithm to improve recommendations for preventive services: comparing algorithm recommendations with those of primary care providers" In press at *Medical Decision Making*
3. Zaaqoq AM, Fox AD, **Boonstra PS**, Riddle C, Chen WT, Priest JR, Hyer S, Rycus P, Stead CM, and Tonna JE (2026) "The Extracorporeal Life Support Organization (ELSO) Registry Data Quality and Integrity Program" *ASAIO Journal* 72:111–116 `https://doi.org/10.1097/MAT.0000000000002660`
PMC 12829488
4. Wang C, Abdelrahman S, Geng X, Burgess A, Hu YS, Rauf MA, Kady N, Fu Y, Reilly TA, Maine IP, **Boonstra PS**, Sabitov K#, Murga-Zamalloa C and Wilcox RA (2026). "CDK9 is a dependency in GATA-3 driven and MCL-1 independent T-cell Lymphomas" *Blood Cancer Journal* 16: Article 9 `https://doi.org/10.1038/s41408-025-01427-1`
#Biostatistics graduate student mentee
PMC 12783670
5. Chase EC#, **Boonstra PS**, and Taylor JMG (2025) "A multiple imputation approach for cumulative incidence estimation" (2025) *The American Statistician* 79:291–301 `https://doi.org/10.1080/00031305.2025.2453674`

#Biostatistics graduate student mentee
PMC 12360614

6. **Boonstra PS** and Orozco del Pino P# (2025) "A comparison of some existing and novel methods for integrating historical models to improve estimation of coefficients in logistic regression" *Journal of the Royal Statistical Society: Series A* 188:46–67 https://doi.org/10.1093/jrsssa/qnae093
#Biostatistics graduate student mentee
PMC 11728056

7. Alexander PM, ..., Barbaro RP for the endorsing societies of **Extracorporeal Life Support Organization (ELSO)** ... (2024) "Definitions of adverse events associated with extracorporeal membrane oxygenation in children: results of an international Delphi process from the ECMO-CENTRAL ARC" *The Lancet Child & Adolescent Health* 8:773–780 https://doi.org/10.1016/S2352-4642(24)00132-9
PMC 12917940

8. **Boonstra PS**, Owen DR#, and Kang J (2024) "Shrinkage priors for isotonic probability vectors and binary data modeling" *Pharmaceutical Statistics* 23:540–556 https://doi.org/10.1002/pst.2372
#Graduate student mentee
PMC 11737611

9. Strohbehn GW, Stadler WM, **Boonstra PS**, and Ratain MJ (2024) "Optimizing the doses of cancer drugs after usual dose-finding" *Clinical Trials* 21:340–349 https://doi.org/10.1177/17407745231213882

10. Tonna JE, **Boonstra PS**, MacLaren G, Paden M, Brodie D, Anders M, Hoskote A, Ramanathan K, Hyslop R, Fanning JJ, Rycus P, Stead C, Barrett NA, Mueller T, Gómez RD, Kapoor PM, Fraser JF, Bartlett RH, Alexander PMA, and Barbaro RP, on behalf of the Extracorporeal Life Support Organization Member Centers (2024) "Extracorporeal Life Support Organization Registry International Report 2022: 100,000 survivors" *ASAIO* 70:131–143 https://doi.org/10.1097/mat.0000000000002128
PMC 10962646

11. Chase EC#, Taylor JMG, and **Boonstra PS** (2024) "Modeling data using horseshoe process regression" *Statistics in Medicine* 43: 817–832 https://doi.org/10.1002/sim.9991
#Biostatistics graduate student mentee
PMC 12930000

12. Foucar CE, Foley DH, Aldous J#, Burke PW, Pettit KR, Benitez LL, Perissinotti AJ, Marini BL, **Boonstra PS**, and Bixby D (2024) "Real-World Outcomes with Immunosuppressive Therapy for Aplastic Anemia in Patients Treated at the University of Michigan" *European Journal of Haematology* https://doi.org/10.1111/ejh.14131 112: 424–432 https://doi.org/10.1111/ejh.14131
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13. **Boonstra PS**, Tabarrok A, and Strohbehn GW (2023) "Targeted randomization dose optimization trials enable fractional dosing of scarce drugs" *PLOS One* 18: e0287511 https://doi.org/10.1371/journal.pone.0287511
PMC 10615276

14. Phillips T, Bond D, Takiar R, Kump K, Kandarpa M, **Boonstra PS**, Mayer T, Nachar V, Wilcox R, Carty S, Karimi Y, Nikolovska-Coleska Z, Kaminski R, Herrera A, Maddocks K, Popplewell K, and Danilov A (2023) "Adding venetoclax to lenalidomide and rituximab is safe and effective in patients with untreated mantle cell lymphoma" *Blood Advances* 7, 4518–4527 https://doi.org/10.1182/bloodadvances.2023009992
PMC 10425679

15. Delaney PG, Eisner ZJ, Thullah AH, Turay P, Kpawuru S, **Boonstra PS**, and Raghavendran K (2023) "Evaluating feasibility of a novel mobile emergency medical dispatch tool for lay first responder prehospital response coordination in Sierra Leone: A simulation-based study" *Injury* 54, 5–14 https://doi.org/10.1016/j.injury.2022.10.010

16. Flynn AJ, Beck A, Boisvert P, **Boonstra PS**, Caverly T, Gittlen N, Meng G, Raths B, Taksler G, and Friedman CP (2023) "CBK model composition using paired web services and executable functions: A demonstration for individualizing preventive services" *Learning Health Systems* 7, e10325 https://doi.org/10.1002/lrh2.10325
PMC 10091204

17. Hinton T, Karnak D, Tang M, Jiang R, Luo Y, **Boonstra PS**, Sun Y, Nancarrow DJ, Sandford E, Ray P, Maurino C, Matuszak M, Green MD, Schipper MJ,Yanik GA, Tewari M, El Naqa I, Schonewolf CA, Ten Haken R, Jolly S, Lawrence TS, and Ray D (2022) "Improved prediction of radiation pneumonitis by combining biological and radiobiological parameters using a data-driven Bayesian network analysis" *Translational Oncology* 21, 101428 https://doi.org/10.1016/j.tranon.2022.101428
PMC 9046881

18. Kohne JG, MacLaren G, Cagino L, **Boonstra PS**, Brodie D, and Barbaro RP (2022) "Tracheostomy practices and outcomes in patients with COVID-19 supported by ECMO: An analysis of the ELSO registry" *Critical Care Medicine* 50, 1360–1370 https://doi.org/10.1097/ccm.0000000000005579
PMC 9380152

19. Li P, Taylor JMGT, **Boonstra PS**, Lawrence TS, and Schipper MJ (2022) "Utility based approach in individualized optimal dose selection using machine learning methods" *Statistics in Medicine* 41, 2957–2977 https://doi.org/10.1002/sim.9396
PMC 9233043

20. Barbaro RP*, MacLaren G*, **Boonstra PS**, Combes A, Agerstrand C, Annich G, Diaz R, Fan E, Hryniewicz K, Lorusso R, Paden ML, Stead CM, Swol J, Iwashyna TJ, Slutsky AS, Brodie D, for the Extracorporeal Life Support Organization (2021) "Extracorporeal Membrane Oxygenation for COVID-19: Evolving outcomes from an international registry" *The Lancet* 398, 1230–1238 https://doi.org/10.1016/S0140-6736(21)01960-7
PMC 8480964

21. West BT, Little RJA, Andridge RR, **Boonstra PS**, Ware EB, Pandit A, and Alvarado-Leiton F (2021) "Assessing selection bias in regression coefficients estimated from nonprobability samples with applications to genetics and demographic surveys" *Annals of Applied Statistics* 15, 1556–1581 https://doi.org/10.1214/21-AOAS1453
PMC 8887878

22. **Boonstra PS**, Little RJA, West BT, Andridge RR, and Alvarado-Leiton F (2021) "A simulation study of diagnostics for bias in non-probability samples" *Journal of Official Statistics* 37, 751–769 https://doi.org/10.2478/jos-2021-0033
PMC 8460089

23. Delaney PG, Eisner ZJ, Thullah AH, Muller B, Gan J, Timbo A, Kamara S, Sandy K, **Boonstra PS**, Scott JW, Raghavendran K (2021) "Evaluating a novel Prehospital Emergency Trauma Care Assessment Tool (PETCAT) for low- and middle-income countries: Lay first responders expand prehospital care in Sierra Leone" *World Journal of Surgery* 45, 2370–2377 https://doi.org/10.1007/s00268-021-06140-1

24. **Boonstra PS**, Braun TM, and Chase EC# (2021) "A modular framework for seamless oncology trials" *Clinical Trials* 18, 303–313 https://doi.org/10.1177/1740774520981939
#Biostatistics graduate student mentee
PMC 8382494

25. Salami SS*, Tosoian JJ*, Srinivas N, Jones, Jr TA, Brockman S, Elkhoury FF, Bazzi S, Plouffe KR, Sumida L, Siddiqui J, Liu CJ, Kunju LP, Morgan TM, Natarajan S, **Boonstra PS**, Sumida L, Tomlins SA, Udager AM, Sisk, Jr AE, Marks LS, and and Palapattu GS (2021) "Serial molecular profiling of low-grade prostate cancer to assess tumor upgrading: A longitudinal cohort study" *European Urology* 79, 456–465 https://doi.org/10.1016/j.eururo.2020.06.041
*Equal contribution authors
PMC 7779657

26. Owen DR#, Sun Y, **Boonstra PS**, McFarlane M, Viglianti BL, Balter JM, El Naqa I, Schipper MJ, Schonewolf CA, Ten Haken RK, Kong FM, Jolly S, and Matuszak MM (2021) "Investigating the SPECT dose-function metrics associated with radiation-induced lung toxicity risk in non-small cell lung cancer patients undergoing radiation therapy" *Advances in Radiation Oncology* 6, 100666 https://doi.org/10.1016/j.adro.2021.100666
#Graduate student mentee
PMC8010578

27. **Boonstra PS**, Ahmed A, Merrill SA, and Wilcox RA (2021) "Ruxolitinib in adult patients with secondary hemophagocytic lymphohistiocytosis" *American Journal of Hematology* 96: E103-E105 https://doi.org/10.

1002/ajh.26091
PMC 8968463

28. Little RJA, West BT, **Boonstra PS**, and Hu J (2020) "Measures of the degree of departure from ignorable sample selection" *Journal of Survey Statistics and Methodology* 8, 932–964 https://doi.org/10.1093/jssam/smz023
PMC 7750890

29. Barbaro RP*, MacLaren G*, **Boonstra PS**, Iwashyna TJ, Slutsky AS, Fan E, Paden ML, Ogino MT, Bartlett RH, Tonna J, Hyslop R, Fanning JJ, Rycus PT, Hyer S, Anders M, Agerstrand C, Hryniewicz K, Diaz R, Lorusso R, Combes A, Brodie D, for the Extracorporeal Life Support Organization (2020) "Extracorporeal membrane oxygenation support in COVID-19: The registry report from the Extracorporeal Life Support Organization" *The Lancet* 396, 1071–1078 https://doi.org/10.1016/S0140-6736(20)32008-0
*Equal contribution authors
PMC 7518880

30. **Boonstra PS** and Krauss JC (2020) "Inferring a consensus problem list using penalized multistage models for ordered data" *Annals of Applied Statistics* 14, 1557–1580 https://doi.org/10.1214/20-AOAS1361
PMC 8345315

31. Baker LH, **Boonstra PS**, Reinke D, Peregrine Antalis E, Zebrack B, and Weinberg RL "Burden of chronic diseases among sarcoma survivors treated with anthracycline chemotherapy: Results from an observational study" (2020) *Journal of Cancer Metastasis and Treatment* 6, 24 https://doi.org/10.20517/2394-4722.2020.36
PMC 8513741

32. Bankhead A, McMaster T, **Boonstra PS**, and Palmbos PL (2020) "Isoform-specific TP63 expression is linked with distinct clinical outcomes in bladder cancer" *EBioMedicine* 51, 102561 https://dx.doi.org/10.1016%2Fj.ebiom.2019.11.022
PMC 6953644

33. Hanks JE, Kovatch KJ, Ali SA, Roberts E#, Durham AB, Smith JD, Bradford CR, Malloy KM, **Boonstra PS**, and McLean SA (2020) "Long-term outcomes, prognostic value, and accuracy of sentinel lymph node biopsy in head and neck melanoma" *Otolaryngology-Head and Neck Surgery* 162, 520–529 https://doi.org/10.1177/0194599819899934
#Biostatistics graduate student mentee
PMC 8012010

34. **Boonstra PS** and Barbaro RP (2020) "Incorporating historical models with adaptive Bayesian updates" *Biostatistics* 21, e47–e64 https://doi.org/10.1093/biostatistics/kxy053
PMC 7868052

35. Ahmed A, Merrill SA, Alsawah F, Bockenstedt P, Campagnaro E, Devata S, Gitlin SD, Kaminski M, Cusick A, Phillips T, Sood S, Talpaz M, Quiery A, **Boonstra PS**, and Wilcox RA (2019) "Ruxolitinib as a targeted therapy in adult patients with secondary HLH" *Lancet Haematology* 6, e630–e637 https://doi.org/10.1016/S2352-3026(19)30156-5
PMC 8054981

36. Andridge R, West BT, Little RJA, **Boonstra PS**, and Alvarado-Leiton F (2019) "Indices of non-ignorable selection bias for proportions estimated from non-probability samples" *Journal of the Royal Statistical Society: Series C* 68, 1465–1483 https://doi.org/10.1111/rssc.12371
PMC 7724611

37. Chase EC# and **Boonstra PS** (2019) "Accounting for established predictors with the multi-step elastic net" *Statistics in Medicine* 38, 4534–4544
#Biostatistics graduate student mentee
PMC 6736688

38. Geer M*, Roberts E*#, Shango M, Till B, Smith S, Abbas H, Hill B, Kaplan J, Barr PM, Caimi PF, Stephens DM, Lin E, Herrera AF, Rosenbaum E, Amengual K, **Boonstra PS**, Devata S, Wilcox RA, Kaminski M, and Phillips T (2019). "Multicenter retrospective study of intravascular DLBCL treated at academic institutions within the United States" *British Journal of Haematology* 186, 255–262 https://doi.org/10.1111/bjh.15923
*Equal contribution authors
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39. Menghrajani K*, **Boonstra PS***, Perkins C, Gowin KL, Mesa RA, Gotlib JR, Wang L, Singer JW, and Talpaz, M (2019). "Predictive models for splenic response to JAK2 inhibitor therapy in patients with myelofibrosis" *Leukemia & Lymphoma* 60, 1036–1042 https://doi.org/10.1080/10428194.2018.1509315
*Equal contribution authors
PMC 6426689

40. **Boonstra PS**, Barbaro RP, and Sen A (2019). "Default priors for the intercept in logistic regressions" *Computational Statistics & Data Analysis* 133, 245-256 https://doi.org/10.1016/j.csda.2018.10.014
PMC 6748335

41. Barbaro RP, **Boonstra PS**, Kuo KW, Selewski DT, Bailly DK, Stone CL, Chow J, Annich GM, Moler FW, and Paden ML (2019). "Evaluating mortality risk adjustment among children receiving extracorporeal support for respiratory failure" *ASAIO Journal* 65, 277–284 https://doi.org/10.1097/mat.0000000000000813

42. Owen DR, **Boonstra PS** , Viglianti BL, Balter JM, Schipper MJ, Jackson W, El Naqa I, Jolly S, Ten Haken RK, and Matuszak MM (2018). "Modeling patient-specific dose-function response using SPECT/CT to personalize prediction of radiation-induced lung toxicity" *International Journal of Radiation Oncology · Biology · Physics* 102, 1265–1275 https://doi.org/10.1016/j.ijrobp.2018.05.049
PMC 6202237

43. Ma VT, **Boonstra PS**, Menghrajani K, Perkins C, Gowin KL, Mesa RA, Gotlib JR, and Talpaz, M (2018). "Treatment with JAK inhibitors in myelofibrosis patients nullifies the prognostic impact of unfavorable cytogenetics" *Clinical Lymphoma, Myeloma, and Leukemia* 18, e201–e210.
PMC 5927833

44. Hawkins PG*, **Boonstra PS***, Hobson ST, Hayman JA, Ten Haken RK, Matuszak MM, Stanton P, Kalemkerian GP, Lawrence TS, Schipper MJ, and Jolly S (2018). "Prediction of radiation esophagitis in non-small cell lung cancer using clinical factors, dosimetric parameters, and pretreatment cytokine levels." *Translational Oncology* 11, 102–108.
*Equal contribution authors
PMC 6002355

45. **Boonstra PS***, Polk A*, Brown B, Hristov AC, Bailey NG, Lu Y, Kaminski MS, Phillips T, Devata S, Mayer T, and Wilcox RA (2017). "A single center phase II study of ixazomib in patients with relapsed or refractory cutaneous or peripheral T-cell lymphomas" *American Journal of Hematology* 92, 1287–1294.
*Equal contribution authors
PMC 6116510

46. Grivas PD, Devata S, Khoriaty R, **Boonstra PS**, Ruch J, McDonnell K, Hernandez LA, Wilfong J, Smerage J, Ison MG, Eisenberg JNS, Silveira M, Cooney KA, and Worden FP (2017). "Low-cost intervention to increase influenza vaccination rates at a comprehensive cancer center" *Journal of Cancer Education* 32, 871–877.

47. von Salomé J, **Boonstra PS**, Karimi M, Aravidis C, Silander G, Stenmark-Askmalm M, Nilbert M, Lindblom A, and Lagerstedt-Robinson K (2017). "Genetic anticipation in Swedish Lynch syndrome families" *PLOS Genetics* 13, e1007012.
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48. Hawkins PG*, **Boonstra PS***, Ten Haken RK, Matuszak MM, Kong FM, Hearn JM, Lawrence TS, Schipper MJ, and Jolly S (2017). "Radiation-induced lung toxicity in non-small-cell lung cancer: Understanding the interactions of clinical factors and cytokines with the dose-toxicity relationship" *Radiotherapy and Oncology* 125, 66–72.
PMC 5645039
*Equal contribution authors

49. Barbaro RP, **Boonstra PS**, Moler FW, Paden ML, Davis MM, Prosser LA (2017). "Hospital-level variation in inpatient costs among children receiving extracorporeal membrane oxygenation" *Perfusion* 32, 538–546.

50. DaSilva AF, Nascimento TD, Jassar H, Heffernan J, Toback RL, Lucas S, DosSantos MF, Bellile EL, **Boonstra PS**, Taylor JMG, Casey KL, Koeppe RA, Smith YR, and Zubieta JK (2017) "Dopamine D2/D3 imbalance during migraine attack and allodynia in vivo" *Neurology* 88, 1634–1641.
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51. Moyer JS, Flynn S, **Boonstra PS**, Kraft C, Chinn S, Baker SR, Schwartz JL, Bichakjian, CK, Fullen D, Durham AB, Lowe L, Johnson TM (2017). "Efficacy of staged excision with permanent section margin control for cutaneous head and neck melanoma" *JAMA Dermatology* 153, 282–288.

52. Soni PD*, **Boonstra PS***, Schipper MJ, Bazzi L, Dess RT, Matuszak MM, Kong FM, Hayman JA, Ten Haken RK, Lawrence TS, Kalemkerian GP, and Jolly S (2017) "Lower incidence of esophagitis in the elderly undergoing definitive radiation therapy for lung cancer" *Journal of Thoracic Oncology* 12, 539–546.
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*Equal contribution authors

53. **Boonstra PS**, Braun TM, Taylor JMG, Kidwell KM, Bellile EL, Daignault S, Zhao L, Griffith KA, Lawrence TS, Kalemkerian GP, and Schipper MJ (2017). "Statistical controversies in cancer research: Building the bridge to phase II: Efficacy estimation in dose-expansion cohorts" *Annals of Oncology* 28, 1427–1435.
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54. Polk A, Lu Y, Wang T, Seymour E, Bailey NG, Singer JW, **Boonstra PS**, Lim MS, Malek S, and Wilcox RA (2016). "Colony-stimulating Factor-1 Receptor is required for nurse-like cell survival in chronic lymphocytic leukemia" *Clinical Cancer Research* 22, 6118–6128.
PMC 5161678

55. **Boonstra PS**, Taylor JMG and Mukherjee B (2016). "Increasing efficiency for estimating treatment-biomarker interactions with historical data" *Statistical Methods in Medical Research* 25, 2959–2971.
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57. Krauss JC, **Boonstra PS**, Vantsevich A, Friedman CP (2016) "Is the problem list in the eye of the beholder?: An exploration of consistency across physicians" *Journal of the American Medical Informatics Association* 23, 859–865.
PMC 4997039

58. **Boonstra PS**, Taylor JMG, Smolska-Ciszewska B, Behrendt K, Dworzecki T, Gawkowska-Suwinska M, Bialas B, and Suwinski R (2016). "α/β ratio for prostate cancer derived from external beam radiotherapy and brachytherapy boost" *The British Journal of Radiology* 89, 20150957.
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PMC 4724093

60. **Boonstra PS**, Taylor JMG and Mukherjee B (2015). "Adaptive shrinkage via the hyperpenalized EM algorithm" *Statistics in Biosciences* 7, 417–431.
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62. **Boonstra PS**, Shen JC, Taylor JMG, Braun TM, Griffith KA, Daignault S, Kalemkerian GP, Lawrence TS, and Schipper MJ (2015). "A statistical evaluation of dose expansion cohorts in phase I clinical trials" *Journal of the National Cancer Institute* 107, dju429.
PMC 4565529

63. Stenzel SL, Ahn J, **Boonstra PS**, Gruber SB, and Mukherjee B (2015). "The impact of exposure-biased sampling designs on detection of gene-environment interactions in case-control studies with potential exposure misclassification" *European Journal of Epidemiology* 30, 413–423.
PMC 4256150

64. **Boonstra PS***, Bondarenko I*, Park SK, Vokonas PS and Mukherjee B (2014). "Propensity score-based diagnostics for categorical response regression models" *Statistics in Medicine* 33, 455–469.
PMC 3911784
*Equal contribution authors

65. **Boonstra PS**, Mukherjee B and Taylor JMG (2013). "Bayesian shrinkage methods for partially observed data with many predictors" *The Annals of Applied Statistics*, 7, 2272–2292.
PMC 3891514

66. **Boonstra PS**, Taylor JMG and Mukherjee B (2013). "Incorporating auxiliary information for improved prediction in high dimensional datasets: An ensemble of shrinkage approaches" *Biostatistics* 14, 259–272.
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67. **Boonstra PS**, Mukherjee B, Taylor JMG, Nilbert M, Moreno V and Gruber SB (2011). "Bayesian modeling for genetic anticipation in presence of mutational heterogeneity: A case-study in hereditary non-polyposis colorectal cancer (Lynch syndrome)" *Biometrics* 67, 1627–1637.
PMC 3176998

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69. Saran R, Hedgeman E, Plantinga L, Burrows NR, Gillespie BW , Young EW , Coresh J , Pavkov M , Williams D and Powe NR, for **the CKD Surveillance Team** (2010). "Establishing a National Chronic Kidney Disease Surveillance System for the United States" *Clinical Journal of the American Society of Nephrology* 5, 152–161.

Submitted / under revision

- Kohne JG, **Boonstra PS**, Dempsey W, Alexander PMA, Friedman ML, Horvat CM, MacLaren G, Raman L, Rintoul N, Vogel AM, and Barbaro RP. "Impact of Changing Cannulation Strategy on Neurologic Injury in Infants with Respiratory Failure"
- Rali AS, Sarma D, Li H, Urner M, **Boonstra PS**, Alexander PMA, Barbaro RP, Alviar C, Jentzer J, Grandin WE, Bacchetta MD, Lima B, Shah Z, Rycus P, Tonna JE "Prognostic implications of complications during VA ECMO for cardiogenic shock"
- Geng X, Abdelrahman S, Wang C, Kady N, Maine IP, Yang Y, Wang S, Robida A, **Boonstra PS**, Lee J, Reneau J, Murga-Zamalloa C, and Wilcox RA "BCL-xL is a therapeutic vulnerability in T-cell receptor and GATA-3 driven non-ALCL T-cell lymphomas"
- Gutierrez MG, Hemenway N, Wilcox RA, and **Boonstra PS** "Phase II/III trials in relapsed/refractory peripheral T-cell lymphomas: a systematic review and recommendations"
- Wilcox R, Murga-Zamalloa C, Wang C, Perez-Silos V, Zevallos-Morales A, Ehyaee V, Inamdar K, **Boonstra PS**, Wang M, Dai Z, Li G, Dils A, Stone M, Bailey N, Gabali A, Ohe R, Tomomi T, Postigo Mac Dowal M, Montes J, Taxa L, Barrionuevo C, Casavilca-Zambrano S, Culler HF, Reichert C, Lage L, Pereira J, Balakrishna J, Reneau J, Marques-Piubelli ML, Miranda R, Kumar D, Xu M, Smith-Hannah A, Frederiksen J, Ovechko V, Lypiachev K, Brown N, and Lim M "GATA-3 identifies high-grade, immunologically impoverished, and treatment refractory PTCL, NOS"
- Sun SW, Zaaqoq AM, Barbaro RP, Tonna JE, and **Boonstra PS** "A comparison of statistical approaches for analyzing clinical outcomes during extracorporeal membrane oxygenation (ECMO) support using the ELSO Registry"

Conference Abstracts / Posters

1. Ye, JC, **Boonstra PS**, Zonder JA, Pianko M, Anderson LD, Lipe B, Boyer D, Talpaz M, and Campagnaro EL (2026) "Phase 2 Study Using Minimal Residual Disease-Driven Adaptive Strategy in Newly Diagnosed Multiple Myeloma with Upfront Daratumumab-Based Therapy (MAESTRO trial)" (International Myeloma Society, 2026)

2. Gutierrez M, Zamalloa CM, Inamdar K, Stone M, Ehyaee V, **Boonstra PS**, and Wilcox R (2025) "GATA-3 identifies a distinct subset of T-cell prolymphocytic leukemia (T-PLL) and is associated with CCR4 expression" *Blood* 146:1877

3. Phillips TJ, Weiss J, Bond D, Xiao A, Lopedote P, Karimi Y, ..., **Boonstra PS**, Mayer T, and Danilov A (2025) "Long-term follow up of the VALOR (venetoclax, lenalidomide, rituximab) in patients with untreated mantle cell lymphoma" *Hematological Oncology* 43:e135_70093

4. Ye JC, **Boonstra PS**, Campagnaro EL, Pianko M, Kin A, Anderson Jr LD, et al. and Zonder JA (2024) "Phase 2 study using minimal residual disease-driven adaptive strategy in newly diagnosed multiple myeloma with upfront daratumumab-based therapy—first report from the Maestro trial" *Blood* 144:3379

5. Takiar R, Ng WL, Shah E, **Boonstra PS**, Karimi YH, Carty SA, et al. and Phillips TJ (2022) "A multi-center study: outcomes among relapsed/refractory diffuse large B cell lymphoma patients with CD20 loss" *Blood* 140: 9510–9511.

6. Takiar R, **Boonstra PS**, Nachar V, and Phillips T (2022) "ABCL-338 impact of DA-R-EPOCH for high-grade B-cell lymphoma patients at the University of Michigan" *Clinical Lymphoma Myeloma and Leukemia* 22:S373–S374

7. Takiar R, **Boonstra PS**, Karimi Y, Carty S, Wilcox RA, and Phillips TJ (2022) "A real-world experience: outcomes among relapsed/refractory diffuse large B cell lymphoma patients with CD20 loss" *J Clin Oncol* 40:e19570

8. Ye JC, **Boonstra PS**, Boyer DF, Anderson LD, Lipe BC, Kin A, et al. and Talpaz M (2019) "A phase 2 study with minimal residual disease (MRD) driven adaptive strategy in treatment for newly diagnosed multiple myeloma with upfront daratumumab-based therapy" *Blood* 134:3191

Student Mentoring / Advising

PhD Advisor/Co-Advisor

PhD Committee Co-chair: Wei-Hsiang (Sean) Sun — 2028 (anticipated)
Department of Biostatistics, UM

PhD Committee Co-chair: Rachel Tucker Gonzalez — 2027 (anticipated)
Department of Biostatistics, UM

PhD Committee Co-chair: Tian Xie — 2025 (withdrew)
Department of Biostatistics, UM

PhD Committee Co-chair: Elizabeth Chase — 2023
Department of Biostatistics, UM

PhD Committee Member

PhD Committee Member: Sergio Martinez — 2027 (anticipated)
Program in Survey and Data Science, UM

PhD Committee Member: Sarah Ferlito — 2026 (anticipated)
Department of Biostatistics, UM

PhD Committee Member: Nathaniel Osher — 2025
Department of Biostatistics, UM

PhD Committee Member (Cognate): Ramya Naraharisetti — 2023
Department of Epidemiology, UM

PhD Committee Member: Pedro Orozco del Pino — 2022
Department of Biostatistics, UM

PhD Committee Member (Cognate): Ali Rafei — 2021
Survey Methodology Program, UM

PhD Committee Member (Cognate): Daniel Rocky Owen — 2021
Department of Nuclear Engineering & Radiological Sciences, UM

PhD Committee Member: Pin Li — 2020
Department of Biostatistics, UM

PhD Committee Member: Zhichao Sun 2016
Department of Biostatistics, UM

Graduate Student Research Assistant Advisor

Wei-Hsiang (Sean) Sun	2023-2026
Qiyuan Shi	2022-2023
Elizabeth Chase	2021-2022
Selena Scott	2021-2022
Pedro Orozco del Pino	2020-2022

T32 Project Mentor

John Lee ('29 [anticipated]); Nathan Hemenway ('28 [anticipated]); Rachel Tucker Gonzalez ('27 [anticipated]); Elizabeth Chase ('23); Emily Roberts ('21);

MS Academic Advisor

Ishan Ghosh ('27 [anticipated]); Jintong Hong ('26); Kirill Sabitov ('25); Xinwei Wang ('25); Su Xu ('25); Chongwei Shi ('24); Ebo Essilfie-Amoah (withdrew '24); Yahui Zhang ('23); Weizhou Qian ('23); Haisheng Xu ('23); Grace Tiernon ('23); Sahiti Kolli ('21); Zhongzhe Ouyang ('21); Chenyi Yu (withdrew '21); Jane Li ('20); Katherine Li ('20); Chun-Hui Lin ('20); Evan Wiewiora ('20); Michelle Earley ('15); Marco Benedetti ('15)

Software

`isotonicBayes`: `R` package implementing the isotonic probability vectors in Boonstra, Owen, and Kang (2023). Available at `https://github.com/psboonstra/isotonicBayes`

`DoseDeescalation`: `R` package implementing and simulating the dose-deescalation designs as proposed in Boonstra, Tabarrok, and Strohbehn (2023). Available at `https://github.com/psboonstra/DoseDeescalation`

`healthds`: `R` package to accompany the 'Data Science for Health Research' Coursera Specialization. Available at `https://github.com/umich-biostatistics/healthds`

`seamlesssim`: `R` package implementing a simulator for modular seamless oncology trials as proposed in Boonstra, Braun, and Chase (2021). Available at `https://github.com/elizabethchase/seamlesssim`. Co-authored with Elizabeth C. Chase

`adaptBayes`: `R` package implementing the adaptive Bayesian priors proposed in Boonstra and Barbaro (2020). Available at `https://github.com/umich-biostatistics/adaptBayes`. Co-authored with Mike Kleinsasser

Clinical Trials

Completed / Terminated

1. "Open-label, single-center phase II study of MLN9708 (Ixazomib) in patients with relapsed/refractory cutaneous and peripheral T-cell lymphomas" PI: R Wilcox; clinicaltrials.gov: NCT02158975
2. "Phase Ib, open label, single center study of pacritinib in relapsed/refractory lymphoproliferative disorders" PI: R Wilcox; clinicaltrials.gov: NCT03601819
3. "A pilot study of ruxolitinib in secondary hemophagocytic syndrome" PI: R Wilcox; clinicaltrials.gov: NCT02400463

4. "Phase I dose escalation study to determine the maximum tolerated dose of the combination of ruxolitinib and bortezomib in patients with relapsed or refractory lymphoma" PI: T Phillips; clinicaltrials.gov: NCT02613598
5. "A phase I/II study of ruxolitinib in combination with nilotinib in patients with chronic phase CML who have achieved a major molecular remission but not a complete molecular remission on a tyrosine kinase inhibitor alone" PI: P Burke; clinicaltrials.gov: NCT02973711
6. "Phase II study of nivolumab and the antagonistic CSF-1R monoclonal antibody cabiralizumab (BMS-986227) in patients with relapsed/refractory peripheral T cell lymphoma" PI: R Wilcox; clinicaltrials.gov: NCT03927105
7. "Phase I/II study of ixazomib and romidepsin in relapsed/refractory peripheral T-cell lymphoma (PTCL)" PI: S Devata; clinicaltrials.gov: NCT03547700
8. "Phase II multi-center single arm study evaluating the effect of adding bortezomib to ibrutinib in ibrutinib relapsed mantle cell lymphoma" PI: T Phillips; clinicaltrials.gov: NCT03617484
9. "Phase IB dose de-escalation study of the PI3k alpha/delta inhibitor, copanlisib given in combination with the immunotherapeutic agents, nivolumab and rituximab in patients with relapsed/refractory indolent lymphoma LYM17-145" PI: T Phillips; clinicaltrials.gov: NCT04431635
10. "A phase II study using a BFM regimen plus tyrosine kinase inhibitor in adult Philadelphia chromosome-positive acute lymphoblastic leukemia" PI: P Burke; clinicaltrials.gov: NCT04845035

Active / Approved

11. "A Pilot Study of the Addition of Venetoclax to Standard Remission Induction Chemotherapy Fludarabine or Cladribine, Cytarabine, and Granulocyte Colony Stimulating Factor (G-CSF) (FLAG or CLAG) for Frontline Therapy of Secondary AML" PI: D Bixby; clinicaltrials.gov: NCT05780879
12. "A phase II pilot study of metformin therapy in patients with relapsed chronic lymphocytic leukemia and untreated CLL patients with genomic deletion 11q" PI: S Malek; clinicaltrials.gov: NCT01750567
13. "Phase I multi-site study evaluating the MTD, safety and efficacy of the combination venetoclax, lenalidomide and rituximab in patients with previously untreated mantle cell lymphoma" PI: Sano; clinicaltrials.gov: NCT03523975
14. "Phase 2 study with minimal residual disease (MRD) driven adaptive strategy in treatment for newly diagnosed multiple myeloma with upfront daratumumab-based therapy" PI: C Ye; clinicaltrials.gov: NCT04140162
15. "Phase II, open label, multicenter study of pacritinib in relapsed/refractory T-cell lymphoproliferative neoplasms" PI: R Wilcox; clinicaltrials.gov: NCT04858256
16. "A phase II study of pembrolizumab and mogamulizumab in advanced-stage, relapsed/refractory cutaneous T-cell lymphomas" PI: R Wilcox; clinicaltrials.gov: NCT05956041

Funding

Effort is annualized calendar months

Submitted

Current

1R01CA306891 , NIH/NCI (PI Wilcox); $2,026,185 3/11/26 – 2/28/31
Novel Multi-targeted Strategies Targeting T-cell Lymphomas and their Microenvironment
Role: Co-I (0.60 calendar months)

CTSU-2022.116 Wilcox UM MS IIT with HCRN (Merck); $375,645 10/16/23 – 10/11/29
A phase II study of pembrolizumab and mogamulizumab in advanced-stage, relapsed/refractory cutaneous T-cell lymphomas
Role: Co-I (0.06 calendar months)

1R01CA278976, NIH/NCI (PI Wilcox); $2,510,327 1/01/24 – 12/31/28
Notch ligands as oncogenic drivers and therapeutic targets in T-cell lymphomas
Role: Co-I (0.30 calendar months)

Extracorporeal Life Support Organization (PI **Boonstra**); $939,762.00 07/01/22 – 06/30/28
ELSO Registry Contract
Role: PI (4.2 calendar months)

2-P30CA046592, NIH/NCI (PI Fearon); $25,866,261 06/01/12 – 05/31/28
Comprehensive Cancer Center Core Grant II Biostatistics Core
Role: Co-I; Member, Biostatistics Core (0.86 calendar months)

5-R01CA265929 , NIH/NCI (PI Wilcox); $2,085,778 9/01/21 – 8/31/27
Pacritinib in rel/refr T-cell lymphomas
Role: Co-I (0.60 calendar months)

CTSU-2018.056 (Janssen); $1,471,760 10/22/19 – 03/31/27
Phase 2 Study with Minimal Residual Disease (MRD) driven adaptive strategy in treatment for newly diagnosed multiple myeloma with upfront daratumumab-based therapy
Role: Co-I (0.15 calendar months)

Completed

1-R01CA236722, NIH/NCI (PI Wilcox); $1,250,000 12/01/19 – 11/30/24
Notch and GATA-3 as novel therapeutic targets in T-cell lymphomas
Role: Co-I

Clinical Trial (PI Phillips); $307,438 03/03/20 – 3/28/24
Hoosier Cancer Research Network/Bristol-Myers Squibb Company
Phase IB dose de-escalation study of the PI3k alpha/delta inhibitor, Copanlisib given in combination with the immunotherapeutic agents, Nivolumab and Rituximab in patients with relapsed/refractory indolent lymphoma LYM17-145
Role: Biostatistician

Clinical Trial (PI Devata); $294,967 09/24/18 – 08/26/23
Hoosier Cancer Research Network/Millennium Pharmaceuticals, Inc.
Phase I/II study of ixazomib and romidepsin in relapsed/refractory peripheral T-cell lymphoma (PTCL)
Role: Biostatistician

Clinical Trial (PI Ye); $782,241 10/22/19 – 8/11/23
Janssen Scientific
Phase 2 study with Minimal Residual Disease (MRD) driven adaptive strategy in treatment for newly diagnosed multiple myeloma (MM) with upfront daratumumab-based therapy
Role: Biostatistician

2-T32CA083654, NIH/NCI (PI Taylor); $192,805 (FY19) 06/01/12 – 03/31/23
Biostatistics Training In Cancer Research
Role: Program Associate Director

5-R01CA129102, NIH/NCI (PIs Taylor/Ghosh); $1,568,291 01/01/09 – 06/30/22
Statistical Methods for Cancer Biomarkers
Role: Co-I

Clinical Trial (PI Wilcox); $567,436 04/16/19 – 03/31/22
Hoosier Cancer Research Network/Bristol-Myers Squibb Foundation
Phase II study of nivolumab and the antagonistic CSF-1R monoclonal antibody cabiralizumab (BMS-986227) in patients with relapsed/refractory peripheral T cell lymphoma
Role: Biostatistician

Extracorporeal Life Support Organization (PI **Boonstra**); $40,000 10/01/19 – 12/31/21
ELSO Registry Contract
Role: PI

Clinical Trial (PI Phillips); $281,854 12/14/18 – 08/01/21

AbbVie, Inc.
Phase I multi-site study evaluating the MTD, safety and efficacy of the combination venetoclax, lenalidomide and rituximab in patients with previously untreated mantle cell lymphoma
Role: Biostatistician

5-U01DE025633, NIH/NIDCR (PI DaSilva); $282,634 08/01/16 – 07/31/21
Investigation and Modulation of the Mu-Opioid Mechanism in Chronic TMD (in vivo)
Role: Biostatistician

1-R21HD090366, NIH/NICHD (PI West); $275,000 09/23/17 – 08/31/20
Indices of Selection Bias for Non-Probability Samples
Role: Co-I

MICHR Pathway Award (PI Zelner); $50,000 09/01/19 – 08/31/20
Integrated Studies of Vaccine Effectiveness and Failure
CTSA support from grant 5-UL1TR002240, NIH/NCATS
Role: Co-I

5-R01NS094413, NIH/NINDS (PI DaSilva); $361,417 09/30/15 – 08/31/20
Investigation and Modulation of the Central Mu-Opioid
Role: Biostatistician

2-P01CA059827, NIH/NCI (PI TenHaken); $1,784,750 05/15/14 – 04/30/19
Optimization of High Dose Conformal Therapy
Role: Co-I

MICHR Pilot Grant Program (PI **Boonstra**); $15,000 03/01/18 – 02/28/19
Measuring consensus in the problem list using penalized regression models for rank data
CTSA support from grant 5-UL1TR002240, NIH/NCATS
Role: PI

Research Agreement, CTI BioPharma Corp. (PI Talpaz); $75,032 12/11/15 – 12/10/17
A New Prognostic Model for Response in Myelofibrosis Patients Treated with JAK2 Inhibitors
Role: Co-I

R6503-16, Leukemia and Lymphoma Society (PI Wilcox); $270,027 10/01/15 – 09/30/17
Peripheral T-cell Lymphoma, not otherwise specified: The cell of origin as a predictive biomarker and therapeutic target
Role: Biostatistician

Award letter dated 10/22/15, MCubed2.0 Executive Committee 10/22/15 – 04/29/17
(PIs Schipper/**Boonstra**/Matuszak); $60,000
*Identifying treatment*biomarker interactions by combining a large dose-toxicity dataset without biomarkers and a similar but smaller dataset with biomarkers*
Role: Co-PI

Award letter dated 08/21/15, Prostate Cancer Foundation (PI Palapattu); $600,000 09/01/15 – 09/01/17
Testing Targeted NK (TaNK) cell Therapy in Prostate Cancer
Role: Co-I

Award letter dated 05/02/14, Cancer Research Committee, 05/01/14 – 04/30/15
UMCCC (PI **Boonstra**); $25,000
Increasing efficiency for estimating treatment-biomarker interactions with historical data
Role: PI

Seminars and Panels – External

1. *The statistician – numbers that matter* 14th EuroELSO Congress, Dublin, Ireland, May 2026. **Invited Seminar**
2. *Dose-finding in Oncology Clinical Trials* Dordt University, Department of Mathematics & Statistics Speaker Series, Sioux Center, IA (virtual), Nov 2025. **Invited Seminar**
3. *Working with Industry Using the ELSO Registry* 36th Annual ELSO Conference, Oxon Hill, MD, Sept 2025. **Invited Seminar**

4. *Some novel clinical trial designs for dose-finding in oncology* Hope College, Department of Mathematics & Statistics Colloquium Series, Holland, MI, Oct 2024. **Invited Seminar**

5. *Some Challenges in Statistical Modeling in ECLS* 35th Annual ELSO Conference, Detroit, MI, Sept 2024. **Invited Seminar**

6. *Why you should consider a graduate education in biostatistics* Calvin University, Department of Mathematics & Statistics Colloquium, Grand Rapids, MI, Sept 2023. **Invited Seminar**

7. *Targeted randomization dose optimization trials enable fractional dosing of scarce drugs* The 36th New England Statistics Symposium, Boston, MA (virtual presenter), June 2023. **Invited Presentation**

8. *Industry use of the ELSO Registry to Support Device Approval and Surveillance* 33rd Annual ELSO Conference, Boston, MA, Sept 2022. **Invited Presentation**

9. *A comparison of methods for incorporating information from historical prediction models* 43rd Annual Conference of the International Society for Clinical Biostatistics, Newcastle, UK, Aug 2022. **Contributed Talk**

10. *ECMO for COVID-19: Evolving Outcomes from the ELSO Registry* 32nd Annual ELSO Conference (Virtual), Sept 2021. **Invited Panel**

11. *The ELSO Registry on ECMO for COVID-19* 31st Annual ELSO Conference (Virtual), Sept 2020. **Invited Panel**

12. *Statistics is not an easy-bake oven* Hope College, Department of Mathematics & Statistics Colloquium Series, Holland, MI, Mar 2020. **Invited Seminar**

13. *Measures of the degree of departure from ignorable sample selection* Joint Statistical Meetings, Denver, CO, July 2019. **Invited Talk**

14. *A modular framework for seamless oncology trials* Columbia University, Dept. of Biostatistics, New York, NY, Mar 2019. **Invited Seminar**

15. *A default prior for the intercept parameter in logistic regression* Joint Statistical Meetings, Vancouver, BC, Aug 2018. **Topic Contributed Talk**

16. *Incorporating historical models with adaptive Bayesian updates* 2018 WNAR Meeting, Edmonton, AB, June 2018. **Contributed Talk**

17. *Combining nested risk-prediction models* 38th Annual Conference of the International Society for Clinical Biostatistics, Vigo, Spain, July 2017. **Contributed Talk**

18. *A default prior for the intercept in binary-data regression models* 2017 ENAR Spring Meeting, Washington, DC, Mar 2017. **Contributed Talk**

19. *Building the bridge to phase II: Efficacy estimation in dose-expansion cohorts* 5th Early Phase Adaptive Trials Workshop, Politecnico di Torino, Turin, Italy, Oct 2016. **Invited Talk**

20. *Efficient designs for dose-expansion in phase I oncology trials* Fred Hutchinson Cancer Research Center, Seattle, WA, Oct 2015. **Invited Seminar**

21. *Leveraging historical data for improved estimates of treatment-biomarker interactions* 36th Annual Conference of the International Society for Clinical Biostatistics, Utrecht, The Netherlands, Aug 2015. **Contributed Talk**

22. *Strategies for tuning parameter selection in regression problems with many covariates* Fred Hutchinson Cancer Research Center, Seattle, WA, Jan 2015. **Invited Seminar**

23. *A statistical evaluation of expansion cohorts in phase I clinical trials* 4th Early Phase Adaptive Trials Workshop, Medical University of South Carolina, Charleston, SC, Oct 2014. **Invited Talk**

24. *Increasing efficiency for estimating treatment-biomarker interactions with historical data* 2014 Midwest Biopharmaceutical Statistics Workshop, Ball State University, Muncie, IN, May 2014. **Contributed Talk**

25. *Prediction with high-dimensional data: A brief foray into biostatistics* Hope College, Department of Mathematics Colloquium Series, Holland, MI, Oct 2013. **Invited Seminar**

26. *Adaptive shrinkage via the hyperpenalized EM algorithm* 34th Annual Conference of the International Society for Clinical Biostatistics, Munich, Germany, Aug 2013. **Contributed Talk**

27. *Adaptive shrinkage via the hyperpenalized EM algorithm* Joint Statistical Meetings, Montreal, QC, Aug 2013. **Invited Talk**

28. *Incorporating auxiliary information for improved prediction in high dimensional datasets: An ensemble of shrinkage approaches* Joint Statistical Meetings, San Diego, CA, Aug 2012. **Contributed Talk**

29. *Incorporating auxiliary information for improved prediction in high dimensional datasets: An ensemble of shrinkage approaches* Highlighted Student Talk, Michigan Student Symposium in Statistical Science, Ann Arbor, MI, Apr 2012. **Contributed Talk**

30. *Prediction with high-dimensional data: A brief foray into biostatistics* Calvin College, Department of Mathematics & Statistics Colloquium, Grand Rapids, MI, Dec 2011. **Invited Seminar**

31. *Incorporating auxiliary information for improved prediction in high dimensional datasets: An ensemble of shrinkage approaches* Conference on Risk Assessment and Evaluation of Predictions, Silver Spring, MD, Oct 2011. **Contributed Talk**

32. *Incorporating auxiliary information for improved prediction in high dimensional datasets: An ensemble of shrinkage approaches* Joint Statistical Meetings, Miami Beach, FL, Aug 2011. **Contributed Talk**

33. *Bayesian modeling for genetic anticipation in presence of mutational heterogeneity: A case-study in Lynch Syndrome* Joint Statistical Meetings, Vancouver, BC, Aug 2010. **Contributed Talk**

Seminars and Panels – Internal

1. *Some novel extensions of the horseshoe prior* University of Michigan, Dept. of Biostatistics, Ann Arbor, MI, Jan 2020. **Invited Seminar**

2. *Incorporating historical information with adaptive Bayesian updates* Workshop on Bayes, Fiducial, and Frequentist Paradigm in Data Integration, Machine Learning, and Applications, University of Michigan, Dept. of Biostatistics, Ann Arbor, MI, Nov 2019. **Invited Seminar**

3. *A modular framework for seamless oncology trials* University of Michigan, Dept. of Radiation Oncology Biostatistics Division Seminar, Ann Arbor, MI, Dec 2018. **Invited Seminar**

4. *Modeling consensus among ordered lists using variable selection penalties* Rod Little Lectureship Dinner, Ann Arbor, MI, Sep 2018. **Invited Speed Talk**

5. *Fifth Bayesian, Fiducial, and Frequentist (BFF5) Conference* University of Michigan, Ann Arbor, MI, May 2018. **Invited Panelist**

6. *Combining Bayesian generalized linear models with map-weighting* Biomedical Statistical Modeling Conference, Ann Arbor, MI, June 2017. **Invited Talk**

7. *Incorporating auxiliary information for improved prediction in high dimensional datasets: An ensemble of shrinkage approaches* Department of Biostatistics Seminar, University of Michigan, Ann Arbor, MI, Nov 2012. **Invited Seminar**

8. *Incorporating auxiliary information for improved prediction in high dimensional datasets: An ensemble of shrinkage approaches* Department of Statistics Student Seminar, University of Michigan, Ann Arbor, MI, Oct 2012. **Invited Seminar**

9. *A review of statistical methods for testing genetic anticipation: looking for an answer in Lynch Syndrome* Molecular Epidemiology of Colorectal Cancer Investigators' Meeting, Ann Arbor, MI, Oct 2009 and June 2010.

Workshops and Guest Lectures

1. *Assessment of predictive models* University of Michigan Big Data Summer Institute, Ann Arbor, MI, July 2025. **Guest Lecture / Tutorial**
2. *Clinical Trial Statistics for Clinical Investigators* Hematology-Oncology Fellows, Rogel Cancer Center, July 2025. **Guest Lecture**
3. *Introduction to R markdown* University of Michigan Big Data Summer Institute, Ann Arbor, MI, July 2024. **Guest Lecture**
4. *Assessment of predictive models* University of Michigan Big Data Summer Institute, Ann Arbor, MI, July 2024. **Guest Lecture**
5. *Introduction to R markdown* University of Michigan Big Data Summer Institute, Ann Arbor, MI, July 2023. **Guest Lecture**
6. *Assessment of predictive models* University of Michigan Big Data Summer Institute, Ann Arbor, MI, July 2023. **Guest Lecture**
7. *Introduction to R markdown* University of Michigan Big Data Summer Institute, Ann Arbor, MI, June 2022. **Guest Lecture**
8. *Assessment of predictive models* University of Michigan Big Data Summer Institute, Ann Arbor, MI, June 2022. **Guest Lecture**
9. *Teaching R with rstudio.cloud* University of Michigan, Dept. of Biostatistics, BIOS 803: Pedagogical Methods For Biostatistics Courses, Ann Arbor, MI, Feb 2022. **Guest Lecture**
10. *De-escalation designs for finding the minimum effective dose of a drug* University of Michigan, Dept. of Biostatistics, BIOS 803: Biostatistics in Cancer Seminar, Ann Arbor, MI, Sept 2021. **Guest Lecture**
11. *Introduction to R* University of Michigan, MICHR Bayesian Workshop Series, Ann Arbor, MI, Aug 2019. **Workshop**
    GitHub repo
12. *Introduction to R markdown* University of Michigan Big Data Summer Institute, Ann Arbor, MI, July 2019. **Guest Lecture**
    GitHub repo
13. *Assessment of predictive models* University of Michigan Big Data Summer Institute, Ann Arbor, MI, June 2019. **Guest Lecture**
14. *Expanding your biostatistics toolkit for biomedical research* University of Michigan, Global StatCore Workshop, Ann Arbor, MI, May 2019. **Workshop**
    GitHub repo
15. *A modular framework for seamless oncology trials* University of Michigan, Dept. of Biostatistics, BIOS 803: Biostatistics in Cancer Seminar, Ann Arbor, MI, Nov 2018. **Guest Lecture**
16. *Introductory clinical trials* Hematology-Oncology Fellows, Rogel Cancer Center, July 2018. **Guest Lecture**
17. *Expanding your biostatistics toolkit for biomedical research* University of Michigan, Global StatCore Workshop, Ann Arbor, MI, May 2018. **Workshop**
18. *Building your biostatistics toolbox for cancer research* Dharmais Cancer Hospital, West Jakarta, Indonesia, Aug 2017. **Workshop**
19. *Building the bridge to phase II: Efficacy estimation in dose-expansion cohorts* University of Michigan, Dept. of Biostatistics, BIOS 803: Biostatistics in Cancer Seminar, Ann Arbor, MI, Sep 2016. **Guest Lecture**

20. *Building your biostatistics toolbox: Clinical Trials* St. Pauls Hospital Medical Millennium College, Addis Ababa, Ethiopia, Mar 2016. **Workshop**

21. *Building your biostatistics toolbox: Clinical Trials* Kenya Medical Research Institute, Nairobi, Kenya, Mar 2016. **Workshop**

22. *Adaptive shrinkage via the hyperpenalized EM algorithm* University of Michigan, Dept. of Biostatistics, BIOS 803: Biostatistics in Cancer Seminar, Ann Arbor, MI, Oct 2013. **Guest Lecture**

23. *Bayesian modeling for genetic anticipation in presence of mutational heterogeneity: A case-study in Lynch Syndrome* University of Michigan, Dept. of Biostatistics, BIOS 682: Applied Bayesian Inference, Ann Arbor, MI, Apr 2011. **Guest Lecture**

Service Activities

Internal

Member, Student Recruitment Committee — Fall 2024 – ongoing
Department of Biostatistics, UM

Chair, Curriculum Committee — Fall 2021 – Summer 2023; Fall 2024 – ongoing
Department of Biostatistics, UM

Member, Protocol Review Committee — Winter 2016 – ongoing
Rogel Cancer Center, UM

Reviewer, MICHR Clinical and Translational Science Pilot Award — Fall 2025
Michigan Institute for Clinical and Health Research, UM

Member, Advisory Committee on Academic Programs — Fall 2021 – Summer 2023; Fall 2024 – Summer 2025
School of Public Health, UM

Member, Qualifying Exam Committee — 2015–2016, 2020–2021, 2022–2023
Department of Biostatistics, UM

Member, Cancer Data Science Faculty Search Committee — Fall 2021
Department of Biostatistics, UM

Reviewer, MICHR Pilot Grant Program – Promoting Progress in Statistics Award — Fall 2021, Fall 2018
Michigan Institute for Clinical and Health Research, UM

Statistician, Hem-Onc Fellows' journal club — Feb 2018, Jan 2017, Aug 2015,
Rogel Cancer Center, UM — Feb 2015, Apr 2014, Aug 2013, Mar 2013

Organizing Committee, Biomedical Statistical Modeling Conference — June 2017
Department of Biostatistics, UM

Chair, Seminar Committee — Fall 2014 – Winter 2015
Department of Biostatistics, UM

Faculty Advisor, Student Brownbag Committee — Fall 2014 – Winter 2015
Department of Biostatistics, UM

Member, Bioinformatics/Statistical Genetics Faculty Search Committee — Fall 2010 – Winter 2011
Department of Biostatistics, UM

External

Ad-hoc peer review service

eLife (2026,2025,2024,2023,2022,2021)
The Lancet Haematology (2026,2025,2023,2021,2020)
Journal of the Royal Statistical Society: Series C (JRSSC) (2026,2025)

Biometrics (2026,2021,2019,2018,2017,2016,2015,2012)
Journal of Statistical Computation and Simulation (2025,2024)
Cancer (2025,2020)
JAMA Oncology (2025,2023,2022,2012)
Journal of Official Statistics (2025)
The Lancet Digital Health (2025)
The Lancet Psychiatry (2024,2023,2022,2021,2020,2019)
International Statistical Review (2024,2023,2020,2013)
Journal of the Royal Statistical Society: Series A (JRSSA) (2024,2023)
The Lancet (2024,2022,2018)
Circulation: Cardiovascular Quality and Outcomes (2024,2019,2018)
PLOS One (2024,2019,2018,2014)
Communications in Statistics (2024)
Statistical Methods in Medical Research (2022,2021,2020,2019)
IEEE Transactions on Cybernetics (2022,2014)
Blood (2023)
Heliyon (2023)
Journal of Survey Statistics and Methodology (2023)
Nature Medicine (2023)
Psychological Methods (2023)
Statistics in Medicine (2021,2020,2018,2017,2014,2013)
Scientific Reports (2021,2020)
Harvard Data Science Review (2021)
International Journal of Radiation Oncology · Biology · Physics (2020,2019,2018,2017,2016,2015,2014)
Iranian Journal of Science and Technology, Transactions A: Science (2020,2019)
Annals of Applied Statistics (2020)
BMC Research Notes (2020,2017)
International Journal of Medical Informatics (2020)
JCI Insight (2020)
Journal of the National Cancer Institute (2020)
The Annals of Applied Statistics (2020)
Journal of the American Statistical Association (JASA) (2019,2018)
Clinical Trials (2019,2017,2016)
Pharmaceutical Statistics (2019)
Cancer Cytopathology (2018,2017,2016)
BMJ Open (2018,2017)
Contemporary Clinical Trials Communications (2018)
Journal of Multivariate Analysis (2018)
The American Statistician (2018,2017)
Bayesian Analysis (2017,2015,2014)
British Journal of Radiology (2017)
JAMA Dermatology (2017)
Journal of Statistical Planning & Inference (2017,2013)
Bioinformatics (2016)
BMC Bioinformatics (2016)
American Journal of Clinical Nutrition (2015)
Journal of Agricultural, Biological, and Environmental Statistics (2015,2014,2013)

Electronic Journal of Statistics (2014)
Journal of the American Medical Association (JAMA) (2014)
Scientific World Journal (2014)
Statistica Sinica (2013)
Computational Statistics & Data Analysis (2012)
Genetic Epidemiology (2011)

Other service activities

Member, Data Safety Monitoring Board 2024 – ongoing
ED-LEAD: Emergency Departments Leading the Transformation of Alzheimer's and Dementia Care (clinicaltrials.gov NCT06079203)

Reviewing Editor, eLife journal 2021 – ongoing

AP Statistics Presenter, Saline High School May, 2022–2026, Saline, MI

Reviewer, National Institutes of Health Apr 2026, Virtual
Clinical Oncology (CONC) Study Section, National Cancer Institute

Reviewer, ASA Health Policy Statistics Section Dec 2025
2026 HPSS Student Paper Competition

AP Statistics Presenter, Pioneer High School May 2025, Ann Arbor, MI

Executive Committee Member, International Society for Clinical Biostatistics (ISCB) Jan 2023 – Dec 2024

Session Chair 2024 International Society for Clinical Biostatistics, Thessaloniki, Greece

Biostatistical Reviewer, American Society of Clinical Oncology Feb 2024, Arlington, VA
2024 Grants Selection Committee – Young Investigator Awards, Panel B

Reviewer, ASA Biometrics Section Jan 2024
2024 David P. Byar Young Investigator Award

Oral Session Chair, 44th Annual Conference of the International Society for Clinical Biostatistics, Milan, Italy, Aug 2023

External Reviewer, Graduate School of Social Sciences, Vrije Universiteit Amsterdam Dec 2022
Reviewer for go/no-go assessment of Santiago Gómez-Echeverry's PhD project

Specialist Reviewer, Department of Defense Nov 2022, Virtual
2022 Prostate Cancer Research Program, DoD Congressionally Directed Medical Research Programs

Member, Data Safety Monitoring Board 2017– 2022
mHealth to Improve Blood Pressure Control in Hypertensive African Americans (MI-BP) (clinicaltrials.gov NCT02955537)

Specialist Reviewer, Department of Defense Sep 2021, Virtual
2021 Prostate Cancer Research Program, DoD Congressionally Directed Medical Research Programs

Member, Data Safety Monitoring Board 2019– 2020
Fibromyalgia and Morning Light Treatment study (Fibrolight) (clinicaltrials.gov NCT03794908)

Reviewer, National Institutes of Health Feb 2020, San Diego, CA
Cancer Biomarkers Study Section, National Cancer Institute

Reviewer, AVIESAN and the French National Cancer Institute Aug 2019
Interdisciplinary approaches in oncogenic processes and therapeutic perspectives: Contributions of mathematics and informatics to oncology

Reviewer, National Institutes of Health Apr 2019, Bethesda, MD
HEAL Initiative: Justice Community Opioid Innovation Network (JCOIN) Clinical Research Centers

Reviewer, ASA Biometrics Section Winter 2019
2019 David P. Byar Young Investigator Award

Invited Academic Guest Editor, PLOS One Winter 2019, Fall 2018

Discussant, National Institutes of Health — Mar 2018, Washington, DC
Senator Paul D. Wellstone Muscular Dystrophy Specialized Research Centers (MDSRCs)

JSM Contributed Session Chair — 2014 Joint Statistical Meetings, Boston, MA

JSM Docent — 2014 Joint Statistical Meetings, Boston, MA

JSM Invited Session Chair — 2012 Joint Statistical Meetings, San Diego, CA

JSM Contributed Session Chair — 2011 Joint Statistical Meetings, Miami Beach, FL

Memberships

International Society for Clinical Biostatistics — since 2013

International Biometrics Society (Eastern North American Region) — since 2011

American Statistical Association — since 2010